



Any questions?
Connect with Weijian!



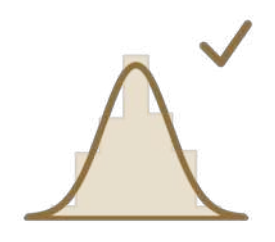
MaRS

Markov-Renewal Single-Photon LiDAR Simulator

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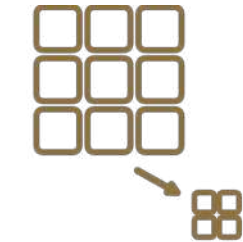


This work was supported in part by CogniSense, one of seven centers in JUMP 2.0, a Semiconductor Research Corporation (SRC) program sponsored by DARPA.



30×

closer to the true photon-count law
KL 0.013 vs 0.409 for the best prior model



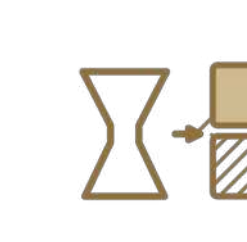
1000×

smaller lookup table
105 MB, not 108 GB — ships with the code



40×

faster than the gold standard
1.01 s per cube — training data at scale



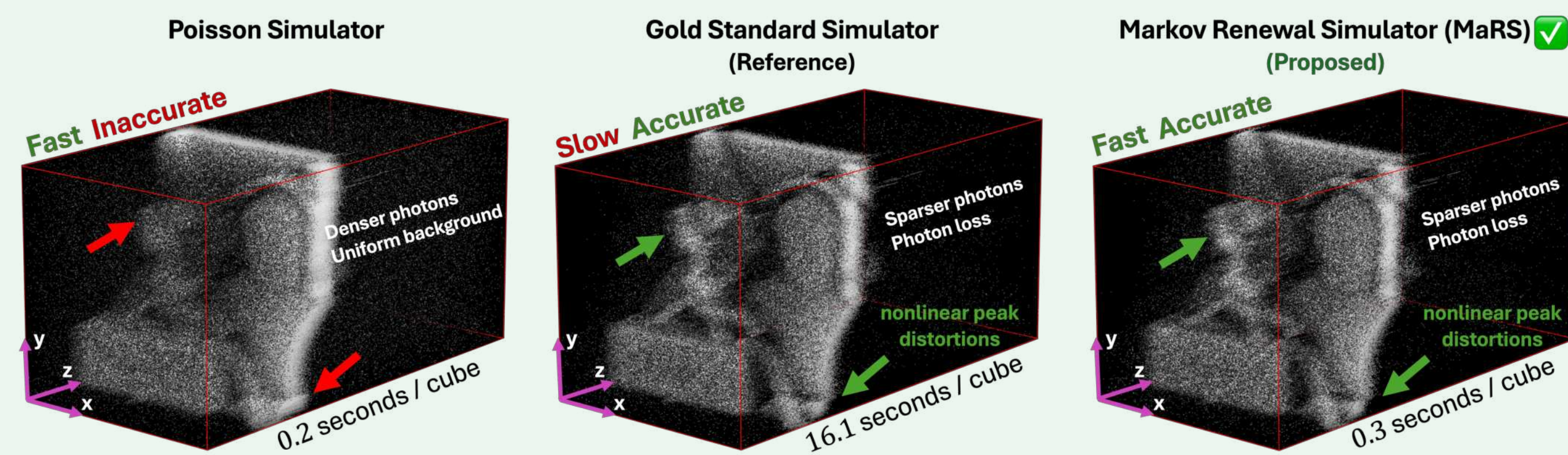
+17 dB

better downstream reconstruction
reflectivity PSNR; depth RMSE 1.65 → 0.07 m

1 The Problem: Fast or Correct?

Single-photon LiDAR builds a 3D histogram of photon arrival times. After each detection, the SPAD is blind for a dead time, so registered photons stop being independent.

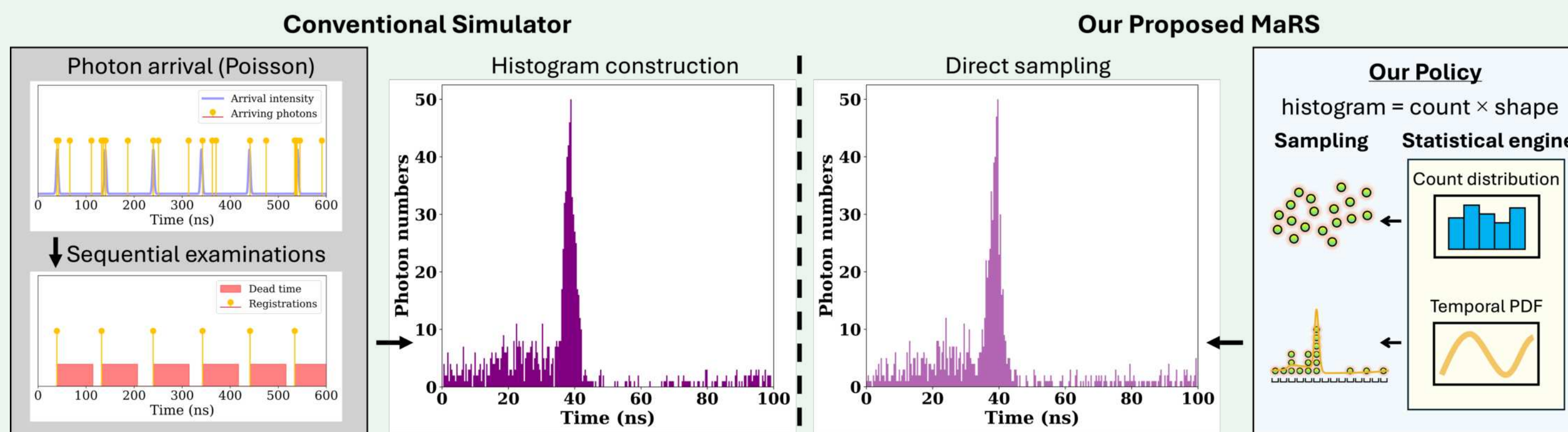
Fast, or physically correct?



Poisson is fast but physically wrong. The gold standard is correct but loops over every photon — it does not scale in photon flux, spatial resolution, or random realizations.

MaRS removes the choice.

2 The Key Idea

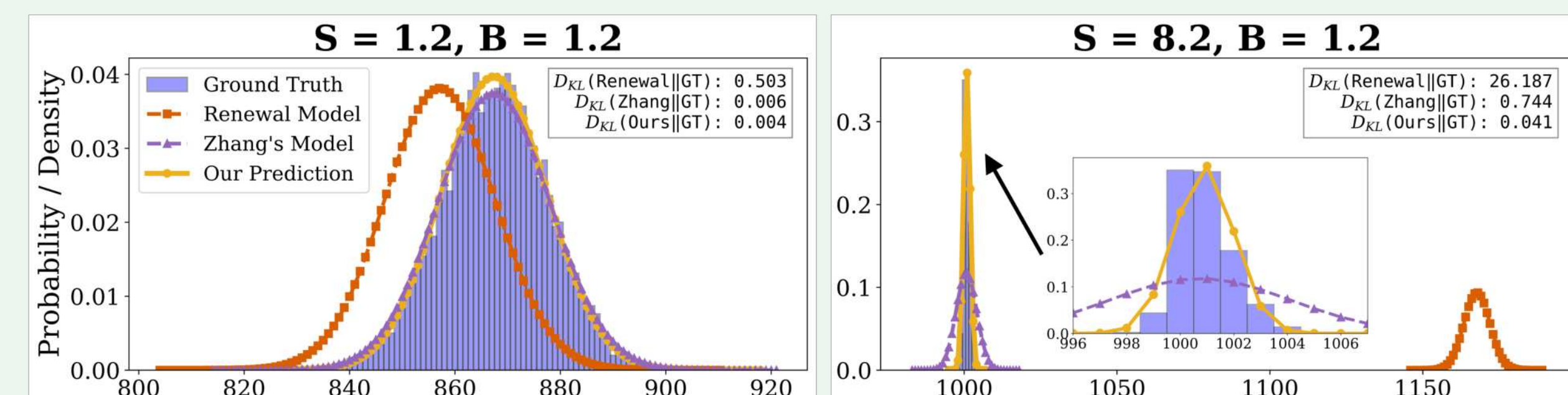


Prior work already predicts the *shape* — where timestamps land. Nobody could predict the *scale* — how many photons the detector actually registers.

MaRS predicts the count.

5 It Is Right — in Simulation and on Real Hardware

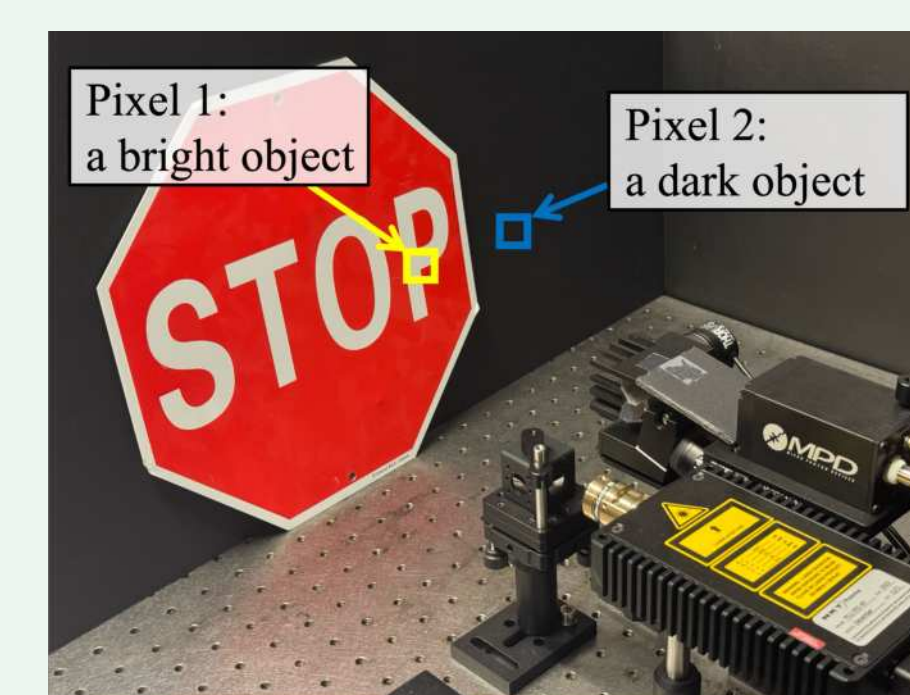
vs. the sequential gold standard



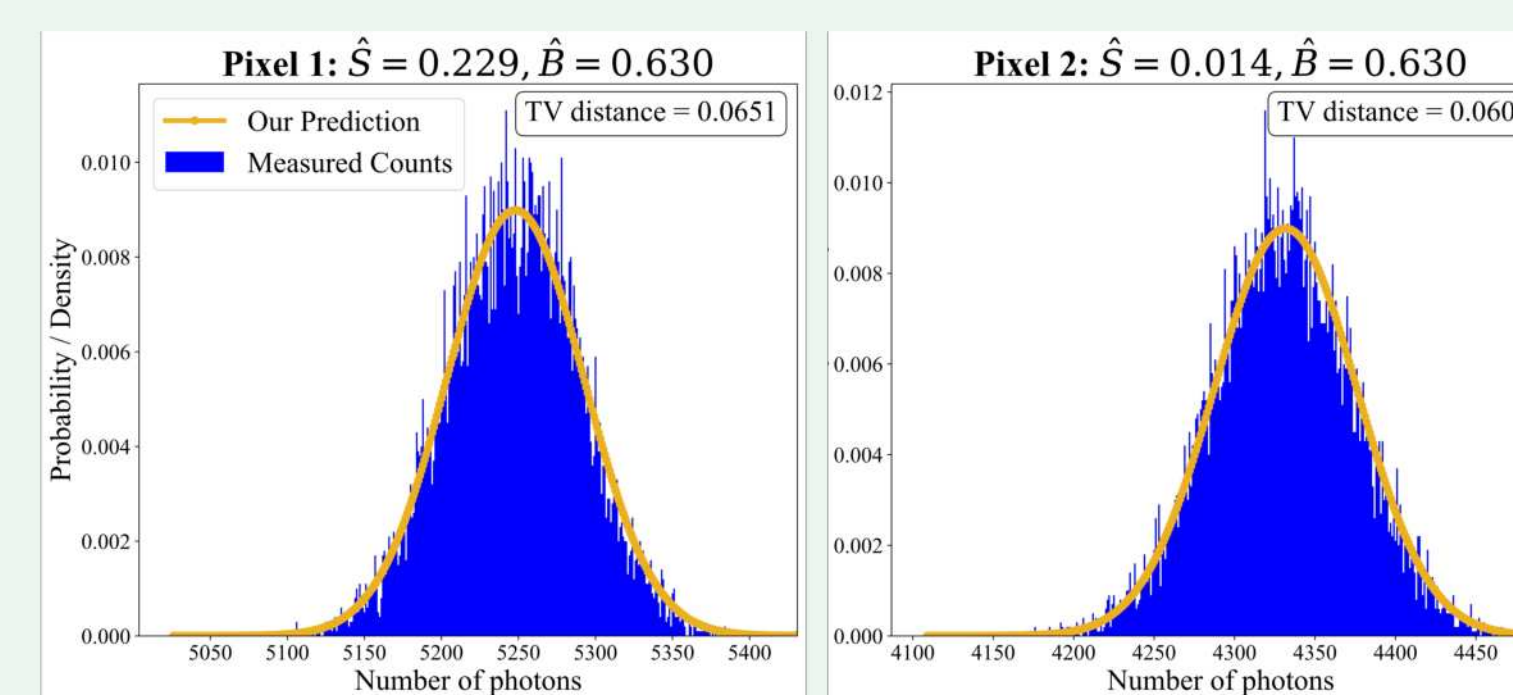
Photon-count distribution vs. sequential ground truth

Simulator	Wasserstein ↓	KL div. ↓
Poisson	7966.13	24.21
Renewal	207.57	18.82
Zhang et al.	2.34	0.409
MaRS (ours)	0.309	0.013

vs. a real SPAD



(a). Experimental setup



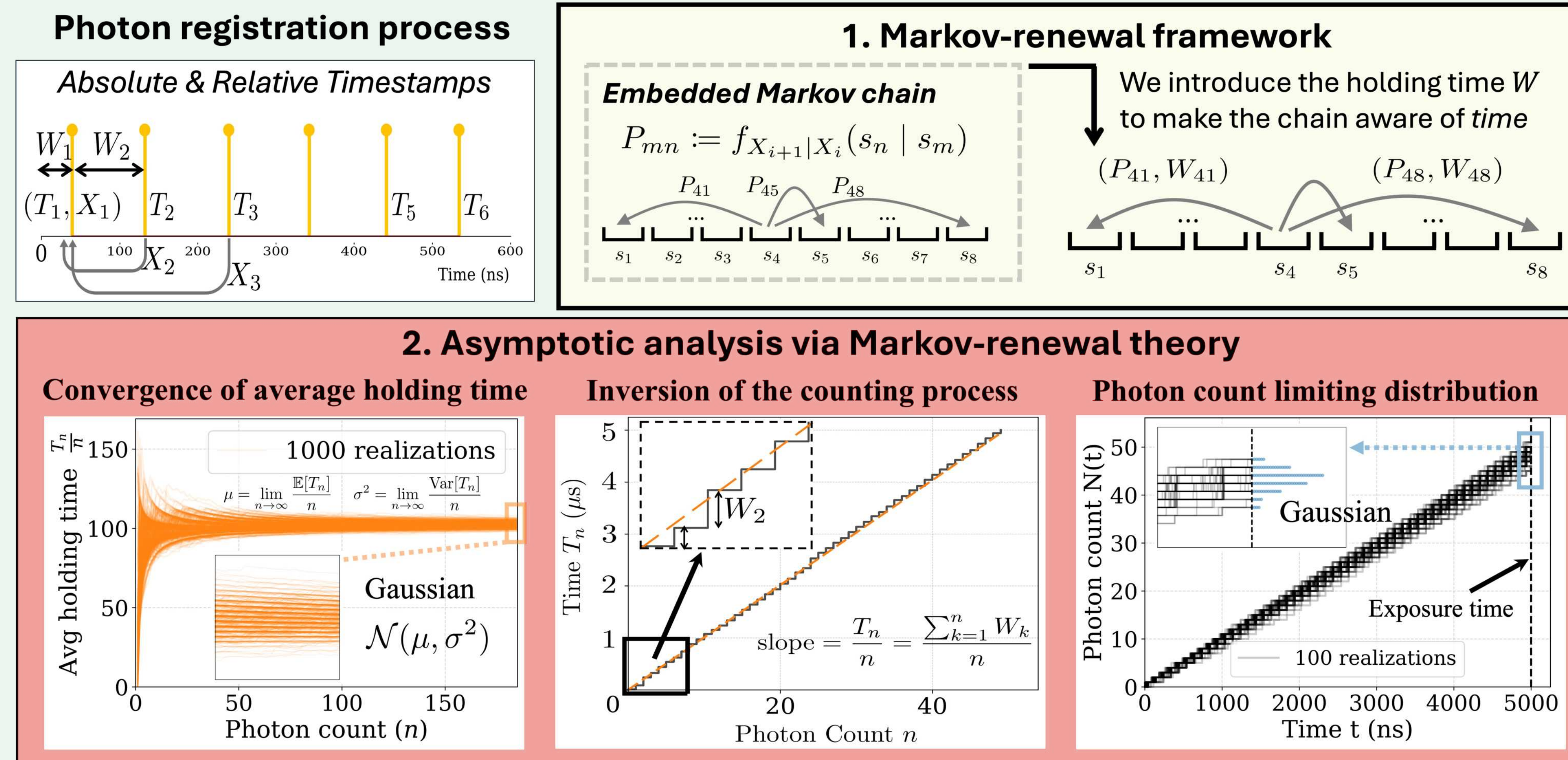
(b). Accuracy of our proposed photon count model

7.5 MHz laser, 85 ns dead time, 10000 realizations. Our predicted count distribution matches the measured one for a bright and a dark target.

Hardware evidence, not just simulator agreement.

3 Counting is the Inverse of Waiting

We model photon registration as a Markov-renewal process: the embedded Markov chain X says where the next photon lands, a holding time W says when. The chain mixes fast, so the cumulative time obeys a central limit theorem (CLT) — inverting it gives the count.



The photon-count law under dead time

$$N(t) \xrightarrow{d} \mathcal{N}\left(\frac{t}{\mu}, \frac{t\sigma^2}{\mu^3}\right)$$

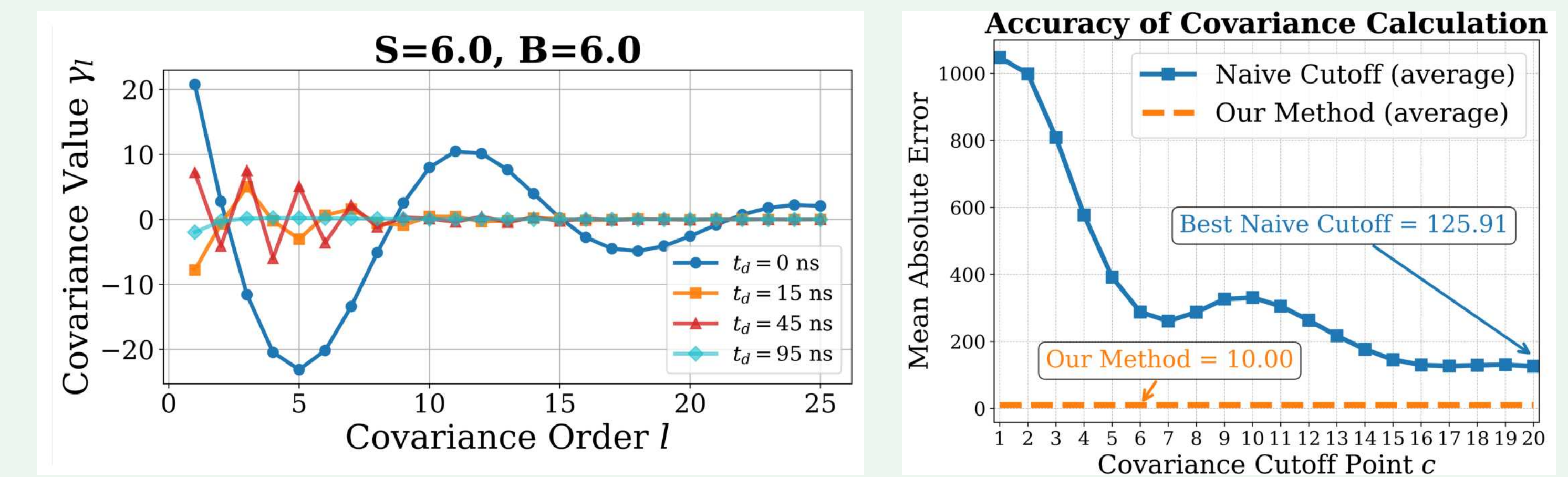
exposure ÷ mean wait

variance from the MRP covariance series

First analytic count law under dead time — no per-photon loop.

4 Scaling Up to Full Histogram Cubes

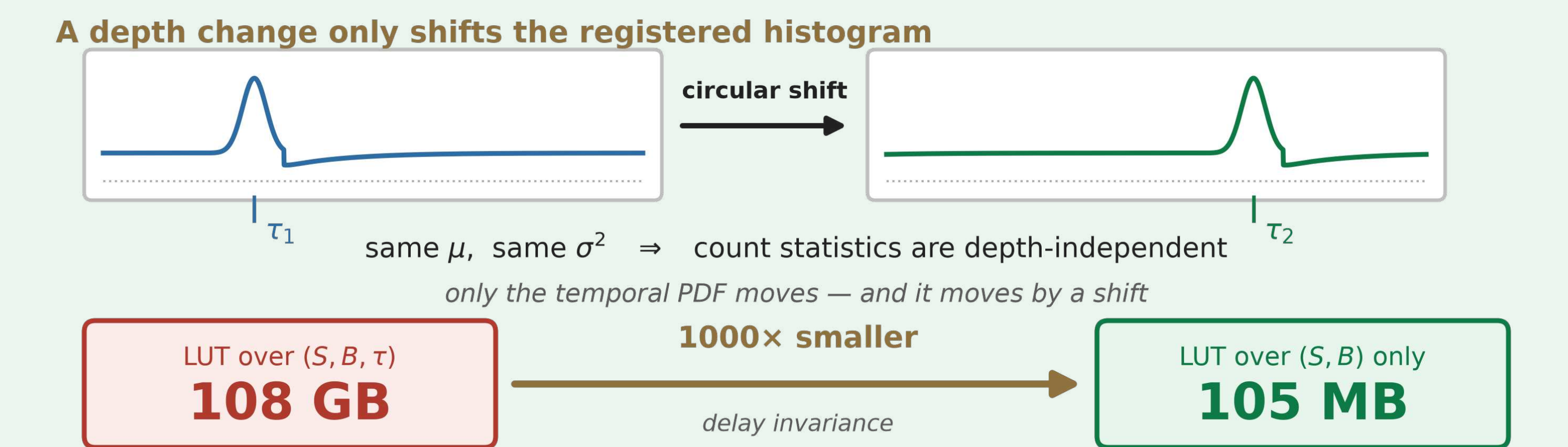
A. Accurate variance under long-lag correlations



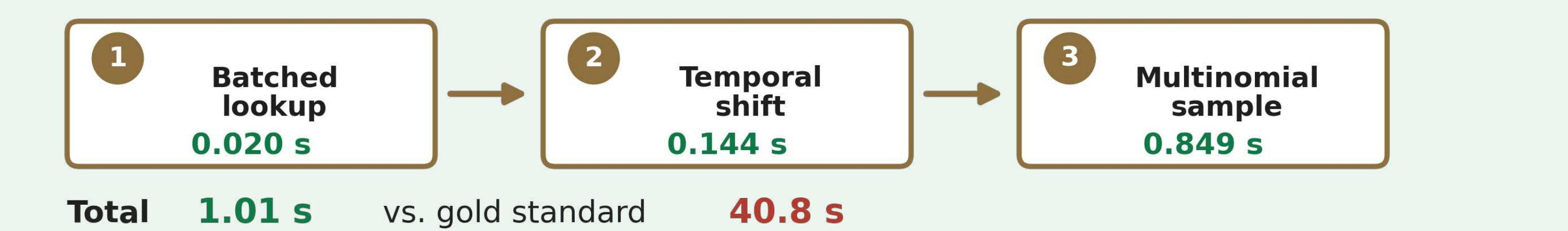
A fixed cutoff is biased or expensive. Our proposed spectral truncation keeps 5 dominant eigenmodes — 12× lower error.

B. A lookup table you can actually ship

Closed-form statistics can be precomputed once. But a table over (S, B, τ) needs 108 GB — τ must span the whole period at fine resolution. Our proved delay invariance theorem removes τ .



Generating a full 128×128×1024 histogram cube



One table per sensor configuration, reused for every pixel and depth.

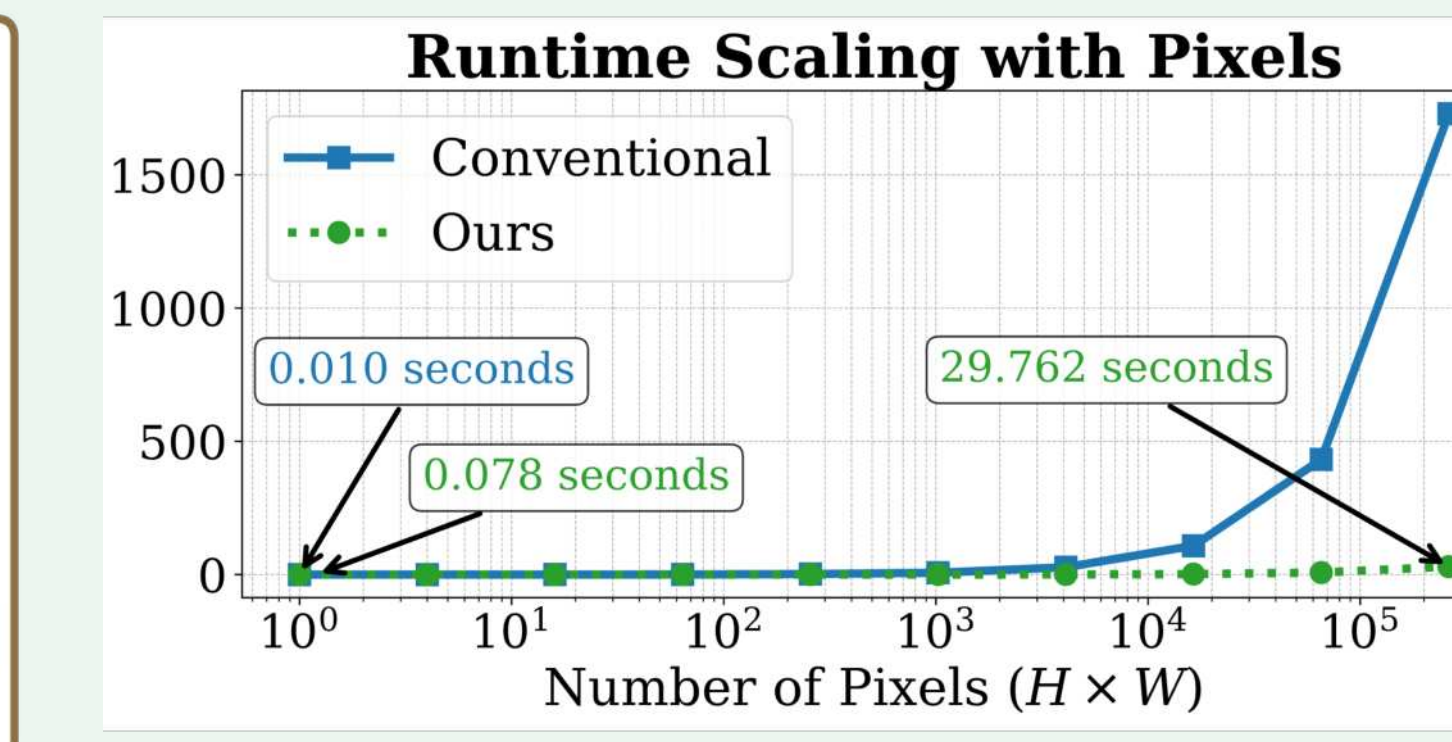
6 Fast Enough to Train On, Faithful Enough to Transfer

Correct statistics or practical speed — until now, not both

dead-time-correct statistics

Simulator	μ	σ^2	π	precomputable
Poisson	X	X	X	
Renewal	X	X	X	✓*
Zhang et al.	✓	✓	✓	✓*
Gold standard	✓	✓	✓	X
MaRS (ours)	✓	✓	✓	✓

* enabled by our lookup-table construction



flat in pixel count — cube generation does not blow up

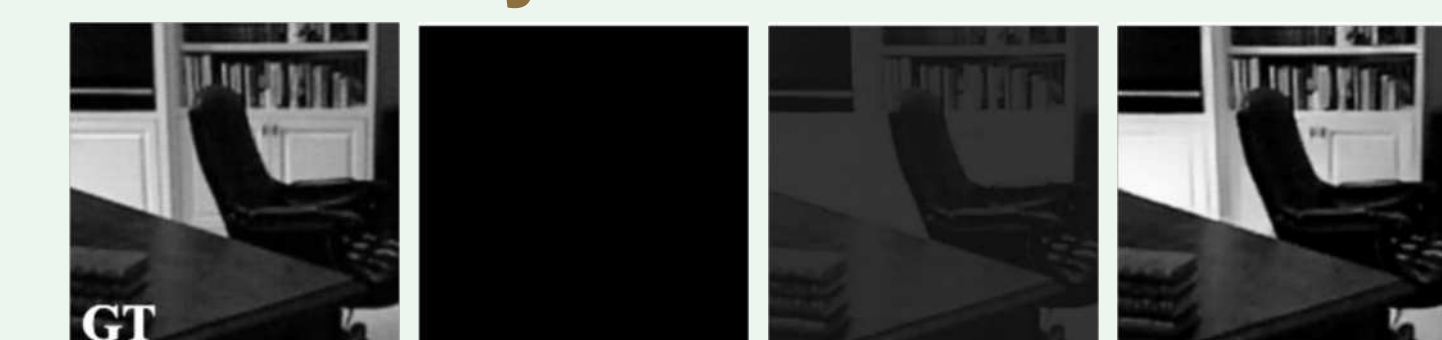
Trained on each simulator, tested on gold-standard physics

● tested on its own simulator ● tested on gold standard

	Reflectivity PSNR (dB) ↑	Depth RMSE (m) ↓, log
Poisson	7.02	0.072
Renewal	12.12	0.054
Zhang et al.	22.74	0.053
MaRS	24.03	0.053

MaRS: smallest gap, best on both

Reflectivity



same network trained on each simulator, all tested on gold-standard data (SBR = 0.6)

Depth



Project page